

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 | SUB-STRAND 1.3: QUADRATIC EQUATIONS | FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems, like building the Harambee hall in Githurai? Use your Driving Question Board, your cumulative model poster and your notes from Lessons 1 to 7. Write your answers in the space under each prompt. Show your working clearly, use the units m, m², KES and s, and check every solution in its context. Round decimals to 2 d.p. in final answers. Write in full sentences where a prompt asks you to explain.

Part 1: Why the committee got stuck

The Githurai committee drew a sketch on manila paper: a rectangular hall with width = w , length = $w + 2$ and Area = 120 m². (a) Write the area as an equation in w and expand it into the form $ax^2 + bx + c = 0$. State a , b and c . (b) The committee also tried the perimeter $2(w + w + 2)$. Write it in simplest form. Compare it with the area equation: which one is linear and which one is quadratic, and why does this explain why the committee could not find w using a linear equation? (c) Look at the trial-and-error table below and explain why guessing is not a reliable method.

w (m)	Length (m)	Area (m ²)
8	10	80
9	11	99
10	12	120
11	13	143
12	14	168

Write your answer here:

Part 2: Expanding and factorising

(a) A Kericho tea farmer fences a plot of length $(x + 5)$ m and width $(x + 3)$ m. Expand the area $(x + 5)(x + 3)$ using the distributive law or FOIL, and write the fencing perimeter in simplest form. Which of the two has an x^2 term, and why? (b) Expand $(2x - 1)(3x + 4)$. (c) Factorise the quadratic expression $w^2 + 2w - 120$ into two brackets. Explain how you found the two numbers you needed, and show how you can check your answer. (d) Explain how expanding and factorising are connected.

Write your answer here:

Part 3: Solving the hall equation three ways

(a) Use the Zero Product Property to solve $(w + 12)(w - 10) = 0$. State the property in your own words. (b) Which solution can be the width of the hall, and why? State the width and length of the hall and check the area. (c) Solve $w^2 + 2w - 120 = 0$ using the quadratic formula. Show the discriminant $D = b^2 - 4ac$. (d) Complete the square on $w^2 + 2w = 120$ to show that the answer is the same. (e) What does the fact that all three methods agree tell you?

Write your answer here:

Part 4: Choosing a method and using context

The Nakuru school garden club has a plot with width x m, length $(x + 6)$ m and area 48 m^2 , so $x^2 + 6x = 48$. (a) Explain why factorisation is not a good method here. (b) Solve it by completing the square and give the width and length to 2 d.p. (c) Mama Wanjiru sells mandazi and wants revenue of KES 700 a day. Her equation is $p^2 - 30p + 175 = 0$, where p is the price in KES. Use the discriminant to find the prices p to 2 d.p. Both prices give the revenue she wants. Which price should she choose, and why? (d) In both problems, how did context help you decide which solution to use?

Write your answer here:

Part 5: Final explanation

Answer the driving question in a short written explanation (8 to 10 sentences): How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems, like building the Harambee hall in Githurai? Your explanation must (1) say how quadratic equations differ from linear ones and why they are necessary, (2) describe how the hall problem was turned into an equation and solved, (3) explain what the discriminant D tells us ($D > 0$, $D = 0$ and $D < 0$) and how context is used to choose a solution, and (4) give one other community problem from Part 4 where quadratic thinking helps, and say how context decided the answer.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Distinguishing quadratic from linear and forming the equation	Correctly writes $w(w + 2) = 120$ as $w^2 + 2w - 120 = 0$, identifies $a = 1$, $b = 2$ and $c = -120$, and clearly explains using $w \times w$ and the linear perimeter $4w + 4$ why a linear method fails.	Forms the equation and identifies a , b and c correctly, with a partial explanation of the difference between quadratic and linear.	Has errors in the equation or in a , b and c , or gives no explanation of why the equation is not linear.
Expanding and factorising accurately	Expands $(x + 5)(x + 3)$ and $(2x - 1)(3x + 4)$ correctly, factorises $w^2 + 2w - 120$ as $(w + 12)(w - 10)$, and checks by expanding.	Expands and factorises mostly correctly, with a minor sign or collecting-terms error or no check.	Several errors in expanding or factorising, and no connection made between the two.
Solving by the three methods	Solves correctly by the Zero Product Property, completing the square and the quadratic formula, including $D = 484$, and	Solves correctly by at least two methods, with minor slips in the working or rounding.	Can solve by one method only, or makes major errors in the formula, the discriminant or the square roots.

	shows the methods agree. Solves the Nakuru and mandazi problems correctly to 2 d.p.		
Using context and the discriminant to judge solutions	Rejects $w = -12$ with a clear reason, states width 10 m and length 12 m with a check of 120 m ² , explains $D > 0$, $D = 0$ and $D < 0$ correctly, and justifies the mandazi price choice.	Rejects the impossible solution and states the answer, with a limited explanation of D or of the price choice.	Gives both solutions without choosing, or gives no reason for rejecting one.
Final explanation and communication	Gives a clear, connected explanation answering the driving question with the hall data, correct terms (solutions, discriminant, Zero Product Property) and another community example.	Explanation covers most required points with mostly correct terms but limited links.	Explanation is incomplete or does not use evidence from the lessons.

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